PROJECT DESIGN PHASE – II

Solution Requirements (Functional & Non-functional)

Project	Global Fuel Prices and Taxation Trends (2020–2026)
Prepared for	Anushka Dubey
Team ID	To be provided
Project period	2020–2026 (data coverage)
Platform	Tableau Public

Project Context

The project, **Global Fuel Prices and Taxation Trends (2020–2026)**, is an interactive Tableau Public dashboard designed to explore and compare petrol, diesel and LPG prices across countries and regions. It helps users examine price movements over time, compare regional and country-level patterns, and understand the estimated tax share in fuel prices.

Dataset scope: 7,056 quarterly records covering 84 countries, 7 regions and the period 2020–2026. The dataset includes country, region, year, quarter, fuel type, price in USD per litre, estimated tax-share percentage and income group.

Functional Requirements

FR No.	Functional requirement	Project-specific requirement / expected behaviour	Priority
FR-1	Data loading	Load the project CSV dataset into Tableau Public and use the fields required for analysis.	High
FR-2	Time-based analysis	Show quarterly fuel-price trends from 2020 to 2026 and allow users to examine changes across years and quarters.	High
FR-3	Fuel comparison	Compare petrol, diesel and LPG prices using a consistent USD-per-litre measure.	High
FR-4	Regional comparison	Compare fuel prices across the 7 regions and identify differences in regional averages and trends.	High
FR-5	Country ranking	Rank or compare countries by fuel price to identify relatively high- and low-price markets.	High
FR-6	Tax analysis	Display estimated tax share (%) and support comparison of taxation patterns across countries, regions and fuel types.	High
FR-7	Interactive filtering	Provide filters for year, region and fuel type so users can focus the dashboard on selected data.	High
FR-8	Dashboard insights	Present KPI cards and four analytical views: Fuel Price Trends, Regional Comparison, Country Ranking and Tax Analysis.	High

Non-functional Requirements

NFR No.	Requirement	Project-specific description
NFR-1	Usability	Dashboard labels, titles, legends and filters should be clear and understandable to a non-technical user.
NFR-2	Performance	Charts and filters should respond smoothly for the supplied dataset of 7,056 records.
NFR-3	Accuracy	Calculations, units and aggregations should be consistent; price values must be identified as USD per litre and tax as an estimated percentage.
NFR-4	Reliability	The workbook should open and display correctly in Tableau Public, with no broken sheets or missing fields.
NFR-5	Maintainability	The CSV source and Tableau worksheets should be organised so that the dataset can be refreshed or the analysis extended.

NFR No.	Requirement	Project-specific description
NFR-6	Transparency	The dashboard must clearly disclose that the dataset contains simulated values and should not be interpreted as official live market prices.

Acceptance criteria: *The dashboard loads the supplied dataset, displays all four analytical views and KPI cards, supports year/region/fuel-type filtering, and clearly communicates the simulated-data disclaimer.*